

Krishnaa Vadivel | Research Engineer | Scientific Computing | Machine Learning | Electrical Engineering

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Summary

Electrical engineering PhD researcher and engineer with experience spanning first-principles materials modeling, semiconductor physics, integrated photonics, scientific machine learning, embedded systems, and performance-oriented software. Combines theory and implementation across Python/C++, CUDA/ROCm, FPGA/MCU platforms, full-stack engineering software, and leadership-class HPC environments.

Experience

Yale University, Electrical Engineering

PhD Research Assistant

2021–present

- Developed graph neural networks and automatic-differentiation models in Python to predict material properties including optical emission spectra.
- Developed first-principles and many-body simulation software in Python and C++ using MPI, OpenMP, PETSc, SLEPc, CUDA, ROCm, and CMake across leadership-class HPC systems.
- Built distributed applications for materials and molecular simulations on Frontier, Frontera, Perlmutter, and related national-lab GPU clusters.
- Developed cleanroom and characterization experience during the early PhD, including GaN power-device fabrication work with MOCVD, selective-area etching, SEM, cathodoluminescence, and Raman analysis.
- Worked on light-matter interaction and quantum many-body modeling, including exciton-polaron and self-trapped-exciton problems relevant to semiconductor and photonic materials.
- Built agentic automation with LangChain, OpenAI API models, and local Ollama models to read documentation and papers, then generate structured inputs for first-principles software.
- Co-authored a 2025 *Journal of the American Chemical Society* paper and delivered APS presentations in 2024, 2025, and 2026 on self-trapped exciton and exciton-polaron research.

Probe Technologies Inc. / Weatherford Wireline Services

Software and Embedded Systems Engineer

2019–2021

- Built parallel applications with CUDA and OpenCL fallback for large sensor-data and waveform processing, diagnostic analytics, and field-data visualization in deployed well-logging systems.
- Developed full-stack server applications with ASP.NET and Microsoft SQL Server for real-time control of company tool testing, data collection, and data processing.
- Developed C# GUI applications for preparing customer inputs and interacting with deployed hardware systems.
- Implemented embedded and digital functionality across Lattice FPGA, PIC32, dsPIC, STM32, and C8051 platforms, including circuit bring-up, oscilloscope validation, and targeted Altium PCB updates.
- Used dolfinx-based finite-element analysis for frequency analysis of steel pipes in wells.

FindLight

Photonics Technical Writer

2018–2019

- Wrote technical content on lasers, optics, and photonics devices for engineering-facing audiences.

Selected Projects

Machine Learning Models for Material Properties: Developed PyTorch Geometric graph neural networks and automatic-differentiation models for molecular-property prediction and optical-properties calculations in material systems.

Integrated Photonics Bandpass Filter: Developed Meep FDTD simulations for a ring-resonator-based integrated-photonics bandpass filter that was later fabricated and tested with an erbium-doped fiber laser.

Container Orchestration for First-Principles Computing: Developed CUDA-ready and cross-environment containers for first-principles software stacks and published them to Docker Hub for portable scientific computing.

Matrix-Multiply AI Accelerator: Developed a Verilog-based matrix-multiply accelerator project for AI-oriented compute pipelines and hardware-software performance exploration.

Education

Yale University

PhD, Electrical Engineering

2021–present

Yale University

MPhil, Electrical Engineering

2023

Yale University

MS, Electrical Engineering

2023

Cornell University

MEng, Electrical and Computer Engineering

2017–2018

Cornell University

BS, Electrical and Computer Engineering

2013–2017

Technical Skills

Programming: Python, C, C++, CUDA, JavaScript, Bash, PowerShell, C#, SQL

Scientific / ML: PyTorch, PyTorch Geometric, scikit-learn, NumPy, pandas, SciPy, SymPy, matplotlib, graph neural networks, transformers

Distributed Compute: MPI, OpenMP, PETSc, SLEPc, mpi4py, petsc4py, slepc4py, ROCm, Frontier, Frontera, Perlmutter

Hardware / Systems: Verilog, Lattice FPGA, PIC32, dsPIC, STM32, C8051, Linux, Docker, Docker Compose, Kubernetes, ASP.NET, Microsoft SQL Server, gcloud

Domains: Computational physics, semiconductor physics, integrated photonics, embedded systems, scientific software